

THE $n = 20$ CASE OF THE INDUCED PLANAR P_6 -TURÁN CONJECTURE

ABSTRACT. Győri and Hama Karim conjectured an exact formula for the induced planar Turán number of the six-vertex path. We verify its first case below Euler's planar bound:

$$\text{exp}(20, P_6^{\text{ind}}) = 53.$$

The lower bound is their construction. The upper bound follows from a computer-assisted census: the unique induced- P_6 -free triangulation on 19 vertices has no induced- P_6 -free facial extension, and no 20-vertex triangulation of minimum degree at least four is induced- P_6 -free.

1. THE EXACT VALUE

All graphs are finite and simple. A triangulation is a plane graph in which every face is bounded by a triangle. For a graph F , let $\text{exp}(n, F^{\text{ind}})$ denote the maximum number of edges in an n -vertex planar graph containing no induced copy of F .

Győri and Hama Karim [4, Conjecture 5.1] conjectured that, for every $n \geq 3$,

$$(1) \quad \text{exp}(n, P_6^{\text{ind}}) = \begin{cases} 3n - 6, & n \leq 19, \\ \left\lfloor \frac{50}{17}(n - 2) \right\rfloor + 1, & n \geq 20. \end{cases}$$

At $n = 20$, the proposed value is 53, whereas Euler's bound is 54.

Theorem 1. *The $n = 20$ case of (1) holds; namely,*

$$\text{exp}(20, P_6^{\text{ind}}) = 53.$$

No claim is made here for $n \geq 21$.

We describe the lower-bound graph without a drawing. Begin with a path $c_0c_1c_2c_3c_4$. Add adjacent vertices a, b , and join each of them to every c_i . For $0 \leq i \leq 3$, insert u_i into the face ac_ic_{i+1} and v_i into the face bc_ic_{i+1} ; then insert w_i into the face $u_ic_ic_{i+1}$. The resulting triangulation T has 19 vertices and 51 edges.

Place a new vertex x in the face abc_0 and add the two edges xa and xb , but not xc_0 . The resulting planar graph W has 20 vertices and 53 edges. It is the $n = 20$ member of the construction in [4, Figure 5 and the preceding paragraph].

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The triangulation T is the one drawn in that figure, which the source asserts, without proof, to contain no induced P_6 ; Proposition 3 below confirms the assertion. The source also observes that repeatedly deleting a degree-three vertex from it yields an induced- P_6 -free triangulation for every smaller n , so the $n \leq 19$ branch of (1) is already settled there.

2. COMPUTATIONAL CENSUS

Let $\mathcal{L}_n$ be a set containing one representative from each isomorphism class of induced- P_6 -free triangulations on n vertices.

Lemma 2. *No simple triangulation of minimum degree at least four on n vertices, $12 \leq n \leq 20$, is induced- P_6 -free.*

The upper bound rests on Lemma 2 twice: it empties the minimum-degree-four branch of the recursion below at every order from 12 to 19, and it is the minimum-degree-four half of $\mathcal{L}_{20} = \emptyset$. Its content is a single sweep. For $12 \leq n \leq 20$ the command `plantri -m4 n` of `plantri` version 5.8 generates 78,435,562 isomorphism classes in total, and the induced-path test described below rejected every one of them. The remaining orders of that branch are small: for $6 \leq n \leq 11$ the same command produces 55 classes altogether. The other branch is small as well: it tests $|\mathcal{L}_{n-1}|(2n-6)$ facial insertions at order n , and 20,196 candidates over $5 \leq n \leq 20$ in all.

Proposition 3. *The graph W is induced- P_6 -free, and*

$$\mathcal{L}_{19} = \{T\}, \quad \mathcal{L}_{20} = \emptyset.$$

More specifically, no 20-vertex triangulation of minimum degree at least four is induced- P_6 -free, and each of the 34 facial insertions of T contains an induced P_6 .

Proof. For each $6 \leq n \leq 20$, the minimum-degree-four branch was generated with `plantri` version 5.8 [3, 1], using

`plantri -m4 n.`

The range starts at 6 because there is no simple triangulation of minimum degree at least four on fewer than six vertices: K_4 has all degrees three, and the unique five-vertex triangulation has two degree-three vertices. Since simple triangulations are 3-connected, the generated plane-isomorphism classes are also abstract isomorphism classes. The runs at orders 19 and 20 produced, respectively,

$$11,284,042 \quad \text{and} \quad 64,719,885$$

triangulations, in agreement with the published enumeration [2, Table 3]. At orders 19 and 20, none passed the induced- P_6 test. Over the whole branch the test was passed only at orders $6 \leq n \leq 10$; for every $11 \leq n \leq 20$ the branch contributed no induced- P_6 -free triangulation at all. This proves Lemma 2.

The remaining triangulations were generated recursively. Every simple triangulation has minimum degree at least three. If it has a degree-three vertex, deleting that vertex leaves a smaller triangulation; conversely, the original graph is recovered by inserting a vertex into a triangular face. Since induced- P_6 -freeness is hereditary under vertex deletion, starting from $\mathcal{L}_4 = \{K_4\}$, adjoining the induced- P_6 -free minimum-degree-four outputs and the induced- P_6 -free facial extensions of members of $\mathcal{L}_{n-1}$, followed by canonical-label isomorphism rejection, yields exactly $\mathcal{L}_n$. The two branches are disjoint, since a triangulation of minimum degree at least four has no degree-three vertex and so is not a facial extension of a smaller triangulation. Consequently $|\mathcal{L}_n|$ is the number of induced- P_6 -free classes in the output of `plantri -m4 n` plus the number of induced- P_6 -free classes obtained by facial extension from $\mathcal{L}_{n-1}$, and by the previous paragraph the first summand vanishes for $11 \leq n \leq 20$. The computation gives

$$(|\mathcal{L}_n|)_{n=4}^{20} = (1, 1, 2, 5, 13, 36, 77, 126, 169, 190, 154, 101, 47, 20, 4, 1, 0).$$

The unique class at order 19 is isomorphic to T .

For the induced-path test, starting from every oriented edge, a partial path $z_1 \dots z_t$ was extended only to a vertex

$$y \in N(z_t) \setminus \{z_1, \dots, z_t\} \quad \text{with} \quad N(y) \cap \{z_1, \dots, z_{t-1}\} = \emptyset.$$

Reaching six vertices is therefore equivalent to finding an induced P_6 . A separate six-subset implementation used the equivalent criterion that the induced graph be connected, have five edges, and have degree multiset $\{1, 1, 2, 2, 2, 2\}$. The two implementations returned identical results. The subset test also verifies directly that W is induced- P_6 -free and supplies an induced path for each of the 34 facial insertions of T . $\square$

Proof of Theorem 1. Proposition 3 gives the lower bound $\exp(20, P_6^{\text{ind}}) \geq 53$ through the graph W .

Suppose that an induced- P_6 -free planar graph G on 20 vertices has 54 edges. Equality in Euler's bound forces G to be a triangulation. If $\delta(G) \geq 4$, this contradicts Proposition 3. Otherwise, G has a degree-three vertex v . Then $G - v$ is an induced- P_6 -free triangulation on 19 vertices, and hence $G - v \cong T$. A simple triangulation is 3-connected and has a unique spherical embedding up to reflection, so G is obtained by inserting v into one of the 34 faces of T . This again contradicts Proposition 3. Therefore 54 edges are impossible, and the upper bound is 53. $\square$

REPRODUCIBILITY

The census used `plantri` version 5.8 [3] and the two implementations of the induced- P_6 test described in the proof of Proposition 3. The source code, the `plantri` source used and its SHA-256 hash, the compilation and execution commands, the logs, and an induced P_6 for each of the 34 facial

insertions of T are to be deposited as ancillary files; none of them is needed for the argument above.

The two graphs the reader may wish to recheck are recorded here directly. In **graph6** format, with the vertices in the order $a, b, c_0, \dots, c_4, u_0, \dots, u_3, v_0, \dots, v_3, w_0, \dots, w_3$, and x last,

R~vMLaWbCKM?U?R?Go@a?EG?KO?KO?

encodes T , and

S~vMLaWbCKM?U?R?Go@a?EG?KO?KOE???

encodes W .

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